

Pre-requisites

1. IBM Cloud Account Setup: [IBM Cloud Console](#)
2. Watsonx Access and Configuration: [Watsonx Overview](#)
3. Python Basics: [Python.org](#)
4. FastAPI Framework: [FastAPI Docs](#)

Project Workflow

1. Gemini AI Setup & Initialization

- **Activity 1.1:** Set up Google Cloud access (or platform) with Gemini 1.5 Flash Pro API access.
- **Activity 1.2:** Configure API keys, usage limits, and test integration for multimodal support (text + image).
- **Activity 1.3:** Fine-tune prompts for budget interpretation and product recommendation generation.

2. Core Functionality Development

- **Activity 2.1:** Design Flask-based backend structure for routing, session control, and planner modules.
- **Activity 2.2:** Create endpoints for Home Planner, Party Planner, and Jewelry Planner.
- **Activity 2.3:** Integrate Gemini prompts and templates for each planner with support for text + optional image inputs.

3. Backend Implementation (app.py)

- **Activity 3.1:** Define Flask routes like `/generate-home`, `/generate-party`, `/generate-jewelry`.
- **Activity 3.2:** Organize backend using modular structure:
 - `routes/`: Endpoints
 - `services/`: Recommendation logic & AI calls
 - `models/`: Input/output schemas
- **Activity 3.3:** Setup CORS headers for frontend communication and implement basic session management.
- **Activity 3.4:** Include mock API calls or simulated scraping from Amazon, IKEA, Zomato, etc.

4. UI Development

- **Activity 4.1:** Build responsive HTML forms for:
 - Home Interior Planner (room details, quantities)
 - Party Planner (event type, budget, guests)
 - Jewelry Planner (budget, occasion, outfit image upload)
- **Activity 4.2:** Add dynamic result display and formatting for recommendations using JavaScript.

5. Testing & Optimization

- **Activity 5.1:** Perform testing with real-world use cases across all three planners.
- **Activity 5.2:** Evaluate Gemini response quality, budget adherence, and platform accuracy.
- **Activity 5.3:** Optimize prompts and add validations to prevent input edge-case failures.
- **Activity 5.4:** Add fallback/default recommendations when AI returns insufficient results.